

FJFI - Department of Physics

02MECH - Mechanics

Test 1

15.11.2023

12:00

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 120 minutes.
2. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
3. Calculators are not allowed. You may only have writing supplies and a transparent water bottle without labels on your desk.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. The maximum score for the test is 4 points.
7. Each question is worth 1 point, but the questions may not require the same amount of time. Therefore, it is advisable to read the entire assignment before starting to solve it.
8. When grading the test, only the solutions provided on the distributed sheets will be considered.
9. Always provide your solutions in separate sheets.

Question 1

Consider two particles moving with velocities described by vectors $\vec{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ m/s and $\vec{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ m/s. At time $t = 0$ s, they were located at points $\vec{r}_{10} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$ m and $\vec{r}_{20} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$ m. Determine the vector of their relative position as a function of time, and calculate time and distance when they are closest to each other.

Solution:

Question 2

You are throwing a ball into the wind, and the wind is blowing in the same direction as the throw of the ball. The wind exerts a constant horizontal acceleration on the ball, represented as $a = \frac{g}{\sqrt{3}}$, where g is the gravitational acceleration. This acceleration acts in the direction of the ball's motion along the x-axis. You initially launch the ball with a velocity of v_0 . Your goal is to find the launch angle α that will allow the ball to travel the farthest horizontal distance possible

Solution:

Question 3

A projectile with a mass m strikes a wall with a velocity v_1 and exits from the other side with a velocity v_2 after penetrating the wall of thickness d . What was the average force F acting on the projectile inside the wall? Use your knowledge of dynamics!

Solution:

Question 4

A mathematical pendulum with a length of l is attached to the arm of a rotating carousel at a distance d from its center. The pendulum is deflected from its vertical position by an angle α . Calculate the period of rotation for the carousel.

Solution: